# Canvas Quiz 4, Modules 6–7a (graded)

Modules 6–7a · Quiz · 5 Oct 2026

Name _____ Date _____

**Question 1** Module 6

Why are multiple dosage regimens are used?

A. to maintain plasma drug levels within the therapeutic window
B. to decrease the MEC
C. to increase the clearance of drugs
D. to decrease the rate of elimination of drug from the body

**Question 2** Module 6

What is the expected maximum steady-state concentration following IV bolus injections of 17 mg/kg to a 75 kg male every 8 hours? The drug has an elimination half-life of 4 hours, and the apparent volume of distribution is 15% of body weight. Give your answer in (mg/L), rounded to the nearest tenth (one decimal place) with no units.

**Question 3** Module 6

Which is the best option to increase the steady-state concentration in a dosing regimen?

A. decrease the dosing interval
B. decrease the elimination half-life
C. increase the clearance
D. increase the dosing interval

**Question 4** Module 6

Four hundred milligrams of an antibiotic was infused intravenously over a period of 2 hours. Eight hours after the start of the first infusion, a second 400-mg dose was infused again over a period of 2 hours. The half-life of the antibiotic is approximately 4 hours and the apparent volume of distribution is approximately 20 L. What is the plasma drug concentration 4 hours after the cessation of the second infusion? Give your answer in mg/L rounded to the nearest tenth (one decimal place) with no units.

**Question 5** Module 6

Which of the following results if the dose is increased on a medication that is to be administered orally at regularly repeating intervals?

A. Decreased clearance
B. Increased steady-state $t_{max}$
C. Decreased rate of absorption
D. Increased steady-state $C_{max}$

**Question 6** Module 6

A 74-kg patient was administered a 750 mg oral dose of a new drug every 8 hours. The drug has an oral bioavailability of 92%, apparent volume of distribution of 0.29 L/kg, elimination half-life of 5 hours, and the absorption half-life of 1.5 hour. When does the expected maximum concentration of drug in the plasma at steady-state occur? Give your answer in hours rounded to the nearest tenth (one decimal place) with no units.

**Question 7** Module 6

Which of the following best describes the assumptions associated with intermittent IV infusions?

A. intermittent IV infusions decrease elimination half-life which leads to increased efficacy
B. intermittent IV infusion is the best way to very rapidly achieve high drug concentrations
C. concentrations of drug in the body following multiple infusions are additive
D. all drugs are better tolerated as IV bolus doses compared to IV infusion

**Question 8** Module 6

A patient is to take a medication by mouth every 8 hours. The medication has an oral bioavailability of 83%, an absorption half-life of 90 minutes, total body clearance of 3.0 L/hr, and apparent volume of distribution of approximately 25 L. What is an appropriate dose to achieve an average plasma drug concentration of 25 mg/L in this patient? Give your answer in mg rounded to the nearest whole number with no units.

## Question 9

Module 7a

What is the absolute bioavailability of a drug for which a 500 mg dose resulted in an AUC of 139 (mg/L)hr? The elimination half-life of the drug is 4.5 hr and the apparent volume of distribution is 18 L. Give your answer as a decimal rounded to the nearest hundredth with no units.

## Question 10

Module 7a

Which term describes the comparison of the bioavailability of the active drug in the systemic circulation following extravascular administration with the bioavailability of the same drug following intravenous administration?

A. Drug Product Performance
B. Relative Bioavailability
C. Bioequivalence
D. Absolute Bioavailability